Performance Comparison of Single- and Dual-Camera Setups for Image-Based Satellite Orbit Determination

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Abstract

This paper aims to determine how well image-based approaches are able to predict satellite trajectories from short observation spans. Four orbit determination methods are presented: two based on a single-camera and two on a dual-camera setup, with one method in each setup determining circular orbits and the other full orbital parameters. The primary experimental setup includes the placement of nine observer cameras in specific locations and variable observation spans. A secondary experimental setup examines the influence of star-based view vector determination on orbit prediction accuracy. Another secondary setup tests a self developed method's ability to determine satellite orbits at different altitudes. Results indicate that methods for full orbit determination predict orbits reasonably well under the right circumstances. However, predicted trajectories are unsuitable for high-precision applications. Methods for circular orbit determination yield more accurate results. Furthermore, we find that there is generally no correlation between view vector determination and orbit prediction, implying that the noise arises from another source. Lastly, we find that the self developed method is indeed applicable to multiple satellite altitudes.
Analysis and series of measurements located here: `https://github.com/stonkchonk/satellite-trajectories`

1 Introduction

Orbit determination of satellites in Low Earth Orbit is achievable by various means. Specifically ground-based methods require measurements spanning long arcs in the range of 3/4 to several resolutions, as stated by Carpenter et al. [1]. This research paper's primary aim is to determine, how well satellite trajectories may be predictable using an image-based approach taking measurements from a single observable flyover, thus short arcs. Hereby, we aim to compare single- and dual-camera setups with fixed camera properties.
This includes a self developed method for circular orbit determination which is based on the initial idea of the Gaussian algorithm for orbit determination. The paper's secondary objective is to determine how susceptible this method is to inaccuracies of star-based view vector determination. Lastly, we aim to determine, whether this new method can predict satellite orbits with differing semi major axes, spanning various altitudes in the Low Earth Orbit region.

2 Methods

2.1 Single-camera orbit determination

2.1.1 Gaussian method

The Gaussian method's approach is to numerically fit observations to a physically feasible orbit. The algorithm yields parameters for full orbit characterization, namely semi major axis a, eccentricity e, argument of periapsis ω and plane normal vector n.
A single observer camera is required, whereas three measurements, each consisting of position vector p_i, view vector v_i and corresponding time stamp t_i need to be recorded to determine a solution. [2]

2.1.2 Simplified Gaussian method

The simplified Gaussian method is a self developed approach to numerically fit observations to a physically feasibly circular orbit. Its incentive stems from the condition that satellite orbits low above earth's surface possess very low eccentricities and can mostly be considered circular. The algorithm yields parameters for circle-orbit characterization, namely semi major axis a and plane normal vector n.
Similar to the Gaussian method, a single observer camera is required, whereas only two measurements, each consisting of position vector p_i, view vector v_i and corresponding time stamp t_i, need to be recorded to de-

termine a solution. Hereby, the solution consists of two scalar parameters ρ_1 and ρ_2 which must satisfy the radius equality condition as well as the arc of a circle condition stated as follows:

$$|r_1| = |r_2| \Leftrightarrow \sqrt{(p_1 + \rho_1 v_1)^2} = \sqrt{(p_2 + \rho_2 v_2)^2} \quad (1)$$

$$\arccos\left(\frac{(p_1 + \rho_1 v_1) \cdot (p_2 + \rho_2 v_2)}{|(p_1 + \rho_1 v_1)|^2}\right) \cdot |(p_1 + \rho_1 v_1)|^{\frac{3}{2}} = (t_2 - t_1)\sqrt{\mu} \quad (2)$$

[3]

2.2 Dual-camera orbit determination

2.2.1 Full parametric trajectory

An orbital trajectory can be described as parametric curve lying in its orbital plane as follows:

$$r(\theta) = \frac{a(1-e^2)}{1 + e \cdot \cos(\theta - \omega)} \cdot \begin{pmatrix}\cos(\theta)\\ \sin(\theta)\end{pmatrix} \quad (3)$$

Given three positions in polar coordinates defined by radii r_i and arguments θ_i, the orbital properties a, e and ω for eccentricities $e \notin \{0, 1\}$ are defined by:

$$\omega = \alpha_{13} - \arctan(G) \quad (4)$$

$$e = \frac{r_1 - r_2}{C_{12} \cdot \cos(\alpha_{12} - \omega)} \quad (5)$$

$$a = \frac{r_1(1 + e \cdot \cos(\theta_1 - \omega))}{1 - e^2} \quad (6)$$

Hereby, G, C_{12}, α_{12} and α_{13} represent terms consisting of r_1, r_2, r_3, θ_1, θ_2 and θ_3.
Two observer cameras are required, utilizing means of stereo vision. Three position vectors of the satellite are obtained by three synchronous measurements which yield values for radii r_i, arguments θ_i and the plane normal vector n. [3]

2.2.2 Circular trajectory

This method once again simplifies the orbit to a circular trajectory. Two observer cameras are required, utilizing means of stereo vision. Two position vectors of the satellite are obtained by two synchronous measurements which yield the semi major axis a and the plane normal vector n. [3]

2.3 Experimental setups

2.3.1 Variable observation spans and observer placement

The goal is to create a measurement setup which can generate data sets usable by all presented single- and dual-camera orbit determination models to evaluate performance under different circumstances. All measurement data is captured from the simulation *Space Engine* which makes the data synthetic. *Space Engine* is a software that incorporates a virtual universe including the solar system and nearby, real stars among other things [4].
Identical virtual cameras are used in all observer locations that generate rectilinear images with the following properties:

Image Property	Value
Image frame [pixels]	1000 x 1000
Field of view [°]	17 x 17

Table 1: Image properties of each observer camera.

In addition, several cameras are placed in strategic locations, so that orbit determination can be performed subject to variable properties.

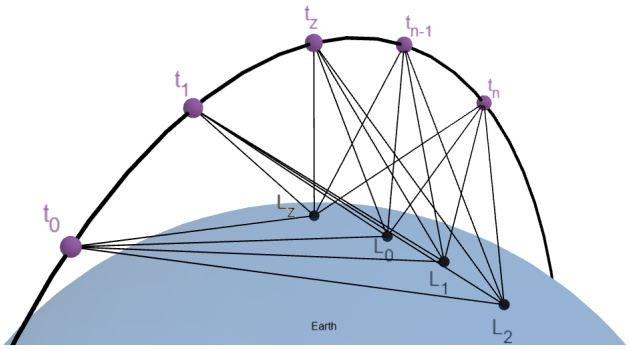


Figure 1: Schematic of strategic camera placement.

The schematic figure above shows the figurative placement of cameras as well as the planned synchronous recording of images. Starting at time stamp t_0 and proceeding with a fixed time interval, each camera takes a picture of the satellite at every time stamp until t_n. For each time stamp from t_0 until t_n and for each observer camera, position vector p_i and view vector v_i are recorded with the Earth Centered Inertial (ECI) coordinate system as reference frame.
As single- and dual-camera orbit determination methods for full determination require only three measurements each and for circular methods only two measurements respectively, we gain some modifiable parameters which can be altered to observe changes in performance and accuracy of the models.
The first one, which is applicable to single- and dual-camera methods, being the alteration of observation time span Δt which translates to the angular passage $\Delta\theta$ that the satellite passes in its orbit. For the Gaussian and stereo vision method for full parametric trajectory determination this is accomplished by spacing apart time stamps by 30 seconds. For the simplified Gaussian and stereo vision method for circular trajectory determination, time stamps are spaced apart by one second.
Secondly, applicable to dual-camera methods, is the variation in distance between two observer cameras which translates to a variable angle between intersecting lines of sight in the context of stereo vision. Consequently, the observer cameras are placed in the following locations on planet Earth within the simulation software:

Location	Latitude	Longitude	Altitude
L_z	$-27.257356°$	$-64.963269°$	313 m
L_0	$-27.301281°$	$-63.939256°$	157 m
L_1	$-27.339896°$	$-63.847839°$	153 m
L_2	$-27.378452°$	$-63.756358°$	155 m
L_3	$-27.455384°$	$-63.573205°$	149 m
L_4	$-27.608528°$	$-63.206133°$	143 m
L_5	$-27.911902°$	$-62.468913°$	127 m
L_6	$-28.506740°$	$-60.982091°$	64 m
L_7	$-29.646743°$	$-57.958463°$	64 m

Table 2: Observation locations

Hereby, it must be noted that given altitude values may deviate from real ones because these were measured within the simulation software. Cameras in locations from L_0 to L_7 are placed in a straight line, whereas the distance on the ground between location $L_{i>0}$ and L_0 doubles for each further location. The distance between L_1 and L_0 equates to 10km, between L_2 and L_0 to 20km and so on up until the distance between L_7 and L_0 equating to 640km. The purpose of location L_z is to observe the satellite passing in the observer's zenith at time stamp t_z. It is not located on line L_0L_7 but close to L_0.
Thirdly, applicable to single-camera methods, is the variation in distance to the orbital plane. Excluding special cases, the distance between observer locations and the orbital plane is constantly changing due to Earth's rotation. However, if the satellite is traversing quick enough from the observer's perspective, an average distance value can be used. Then, a ground truth orbit must be defined, applicable to all presented orbit determination models with the following properties:

Ground Truth Orbit Property	Value
Semi major axis [km]	7067.445
Eccentricity	0
Inclination [°]	52
Ascending Node [°]	0
Reference epoch	J2000
Initial true anomaly θ_0 [°]	0

Table 3: Ground truth orbital properties for most of the experimental setups.

Consequently, the orbit should be circular. It is tilted against the equatorial plane of Earth, whereat the plane normal vector derives from *inclination* and *ascending node* as $n = (0.0, -0.78801075, 0.61566148)$.
Furthermore, it is noted that location L_z was chosen for the predefined time stamp t_z = 2000.01.01 20:03:00 [UT], so that the satellite passes in the observer's zenith with given ground truth properties.

2.3.2 Influence of star-based view vector determination

As elaborated in the preliminary text, view vector determination relies three known stars or rather their view vectors v_{S1}, v_{S2} and v_{S3} to form a matrix M for triangulation [3] which is more or less suited for view vector determination through triangulation. One metric to measure suitability is the determinant d, as it indicates linear dependency of its row/column vectors formulated as follows:

$$d = \det(M) \tag{7}$$

Another useful metric is the condition number $\kappa(M)$ which indicates how sensitive the triangulation matrix M is to errors formulated as follows:

$$\kappa(M) = \|M\|_2 \cdot \left\|M^{-1}\right\|_2 \text{ with } \quad \|M\|_2 = \max_k \sigma_k \tag{8}$$

Hereby, the 2-norm of matrix M is defined by its maximum singular value σ_k [5].
We hypothesize that there might be a correlation between the accuracy of a predicted orbit and the suitability of matrix M used for view vector determination. For testing purposes, we perform orbit determination of the ground truth orbit, defined in table 3, using distinct sets of stars for view vector calibration. Observations are conducted by one camera placed at L_5 starting at certain predefined time stamps so that numerous background stars are visible to form many distinctive sets of stars for calibration. The determination algorithm of choice is the simplified Gaussian method with a fixed observation span $\Delta t = 7s$.

2.3.3 Influence of variable semi major axis

The goal of this setup to verify, whether the self developed simplified Gaussian method holds under differing conditions, specifically with varied semi major axes. Other experiments use solely the ground truth orbit defined in table 3. This experiment modifies the orbit's semi major axis and initial true anomaly θ_0 so that the satellite always passes in the zenith of L_z and t_z which makes it easily visible from all observer locations. The semi major axis is chosen in such a way that the range of mean altitudes cover the entire range of the Low Earth Orbit region as follows:

Semi major axis	Mean altitude	θ_0
6667.445 km	300 km	198.86°
6867.445 km	500 km	282.36°
7067.445 km	700 km	0.00°
7267.445 km	900 km	72.33°
7467.445 km	1100 km	139.85°
7667.445 km	1300 km	203.00°
7867.445 km	1500 km	262.16°
8067.445 km	1700 km	317.67°
8267.445 km	1900 km	9.85°
8467.445 km	2100 km	58.97°

Table 4: Modified properties of ground truth orbit.

Once again, observations are conducted by one camera placed at L_5 with a fixed observation span $\Delta t = 7s$.

2.4 Metrics

2.4.1 Parameter stability

All presented orbit determination methods yield values for semi major axis a and plane normal vector n. Additionally, methods not assuming a circular orbit, yield values for eccentricity e and argument of periapsis ω. The goal is to compare measured (denoted m) with ground truth (denoted g) properties in a generalized form independent of units. For this reason, we define error metrics f_a, f_e, f_ω and f_n for a, e, ω and n as follows::

$$f_a = \frac{|a_g - a_m|}{a_g} \tag{9}$$

$$f_e = |e_g - e_m| \tag{10}$$

$$f_\omega = |\omega_g - \omega_m| \tag{11}$$

$$f_n = \arccos\left(\frac{n_g \cdot n_m}{|n_g| \cdot |n_m|}\right) = \arccos(n_g \cdot n_m) \tag{12}$$

Hereby, specifically for metric 11, it must be noted that this expression is undefined if ground truth or measured orbit has an eccentricity of $e = 0$.

2.4.2 Propagation error

The determined orbit of a satellite differs from the actual ground truth by a margin of error. As the satellite continues orbiting Earth, actual and predicted position continue drifting apart due to this error. Carpenter et al. state that the position error along the trajectory arises from the semi major axis error. This formulates as an equation for the variation in orbital period δT subject to variation in semi major axis δa as:

$$\delta T = 3\pi\sqrt{\frac{a}{\mu}} \cdot \delta a = 3\pi\sqrt{\frac{a}{\mu}} \cdot |a_g - a_m| \tag{13}$$

[1]
Our approach is to formulate the position or rather propagation error in such a way that factors in all orbital parameters and that it can be described as distance metric between the satellite ground truth position s_g and the measured or predicted satellite position s_m.

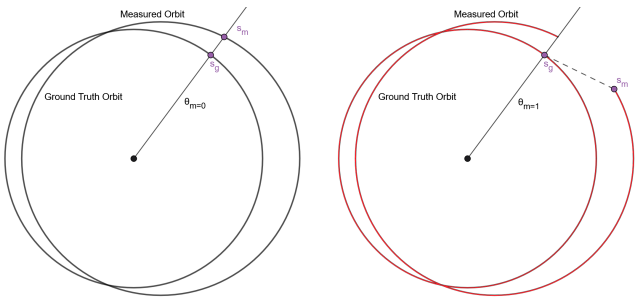


Figure 2: Schematic of propagating error between ground truth and predicted position.

The schematic figure above illustrates how the propagation error accumulates over the course of a single ground truth orbit. The location vector of a satellite can be described subjected to propagated true anomaly θ or passed time t or number of orbital revolutions m since measurement time t_0. Thereby, we define the propagation error as distance between ground truth and predicted position in relation to the ground truth semi major axis a_g subject to orbital revolutions m since t_0 as follows:

$$f_p = \frac{|s_g(m) - s_m(m)|}{a_g} \tag{14}$$

3 Results

In the following, we evaluate error metrics f_a, f_e, f_n and f_p subject to different properties for defined experimental setups. Regarding propagation error f_p, it must be noted that this value is only calculated for plausible predicted orbits. If the predicted orbit has an eccentricity of $e \geq 1$ or has a periapsis below 150km above Earth's surface, it is considered implausible, resulting in the corresponding tile intentionally left blank as visible in figures 6 and 10.
Stereo vision models require measurements from two observer locations denoted as L_iLj with an angle δ which describes the mean intersection angle between lines of sight for all synchronous measurements from t_0 to t_n.
Lastly, it must be noted that no error f_e is determined for circular ground truth to circular model comparison, as it is always 0 in that case. Furthermore, error f_ω is not determined for any experiment, as it is undefined due to circular ground truth orbits.

3.1 Variable observation spans and observer placement

Gaussian method:
Figures 3, 4 and 5 show error values f_a, f_e and f_n subject to single observer location L_i and angular passage $\Delta\theta$. Hereby, prediction stability seems to be dependent on observer location and angular passage. We observe that errors decrease with growing $\Delta\theta$ further from L_z. At $\Delta\theta > 25°$ and ideal observer location, the magnitude of f_a equates to roughly $10^{-2}a_g \mathrel{\widehat{=}} 71km$. At L_z, errors remain persistently high, indicating a numerical instability of the Gaussian method. Plane normal angle deviation seems to converge in broader areas to magnitudes of roughly $f_n = 0.2°$.
The error propagation value f_p for propagated orbits $m = 1$ behaves in a similar manner, as pictured in figure 6. Excluding a few individual tiles, the magnitude of f_p in ideal situations equates to roughly $0.1a_g \mathrel{\widehat{=}} 707km$.

Stereo vision (full parametric model):
Figures 7, 8 and 9 show error values f_a, f_e and f_n subject to intersection angle δ and angular passage $\Delta\theta$. Once again, prediction stability seems to be dependent on both properties, so that accuracy increases with larger δ and $\Delta\theta$. At $\Delta\theta > 25°$ and $\delta_{max} = 25.96°$ the magnitude of f_a equates to roughly $10^{-2.4}a_g \mathrel{\widehat{=}} 28km$. Plane normal angle deviation converges in broader areas to magnitudes of roughly $f_n = 0.1°$.

Lastly, f_p values as pictured in figure 10 contain less plausible tiles. However, under optimal conditions f_p equates to roughly $0.02a_g \mathrel{\widehat{=}} 141km$. Overall, this method yields slightly more accurate results than the Gaussian method.

Simplified Gaussian method:
Figures 11, 12 and 13 show error values f_a, f_n and f_p subject to observer location and angular passage. Error values seem to converge at $\Delta\theta \geq 0.18°$ which equates to an observation span of just $\Delta t \geq 3s$. There does not seem to be an accuracy pattern caused by observer location, rather differences seem to be caused by noise, meaning that location has no notable effect on results. Ideally, error f_a equates to roughly $10^{-4}a_g \mathrel{\widehat{=}} 0.7km$, plane normal deviates by $f_n = 0.1°$ and
$f_p < 10^{-2}a_g \mathrel{\widehat{=}} 71km$. As there is no pattern subject to observer location, these ideal results can not be assumed in every case.
Overall, this method is more accurate than the Gaussian and stereo vision method for full parametric trajectories in determining properties of circular orbits. A particular advantage over the classical Gaussian method is the aspect that there is no numerical instability at L_z and other close locations.

Stereo vision (circular model):
Figures 14, 15 and 16 show error values f_a, f_n and f_p. All three error values decrease with increased intersection angle δ. No change in magnitude can be observed by altering $\Delta\theta$.
Overall, this method is more accurate than the Gaussian and stereo vision method for full parametric trajectories in determining properties of circular orbits. Compared to the simplified Gaussian method, error values are less susceptible to noise. A particular disadvantage is the aspect, that a second camera is needed which needs to be placed at substantial distance from the first one.

3.2 Influence of star-based view vector determination

The Gaussian, stereo vision for full parametric trajectories and simplified Gaussian methods yield noisy error results, hypothesized to originate from star-based view vector determination. Hence, we observe propagation error f_p subject to determinant or condition number and m propagated circulations. Figures 17 and 18 illustrate these relations where the same orbit is predicted exemplary by the simplified Gaussian method, calibrating view vectors on 153 distinct sets of three stars.
We observe that for a wide range of determinants or conditions numbers, f_p remains relatively low for the first few m circulations. As m increases, f_p increases for some calibrations but not for all. There does not seem to be a general clear correlation as f_p for larger m may be low for less favorable determinants or condition numbers.
A steep increase of f_p is only observable in extreme cases $\kappa > 1110$ and $d < 0.000387$.

3.3 Influence of variable semi major axis

Figure 19 shows error f_p subject to observation span Δt and semi major axis a for the simplified Gaussian method.
We observe that it is possible to obtain propagation errors $f_p < 10^{-2}a_g$. This error is sometimes larger but does not appear to dependent on a, rather a result of noise.
This proves that the self developed simplified Gaussian method can be applied to satellites in a variety of altitudes in the Low Earth Orbit region.

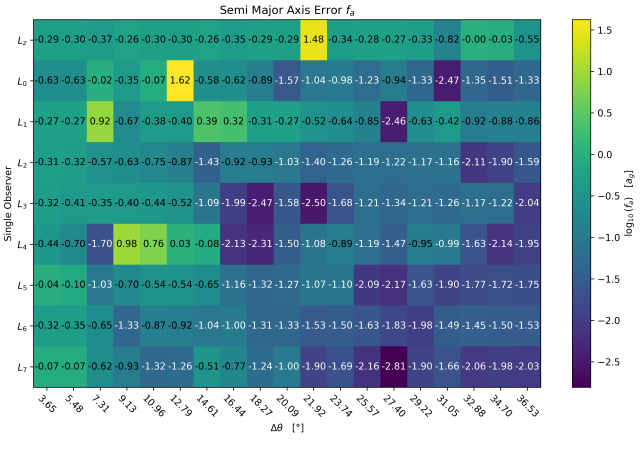


Figure 3: Gaussian method: Stability of semi major axis.

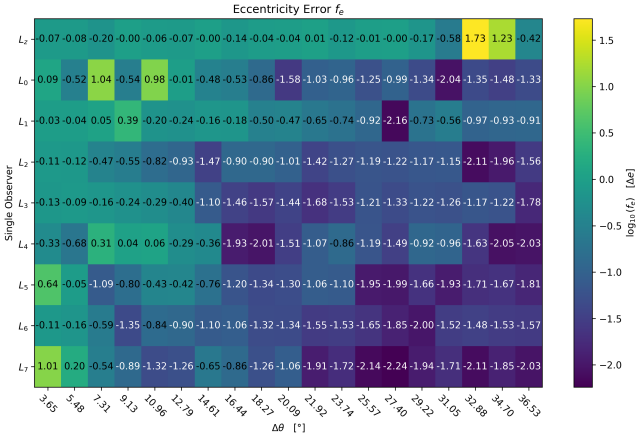


Figure 4: Gaussian method: Stability of eccentricity.

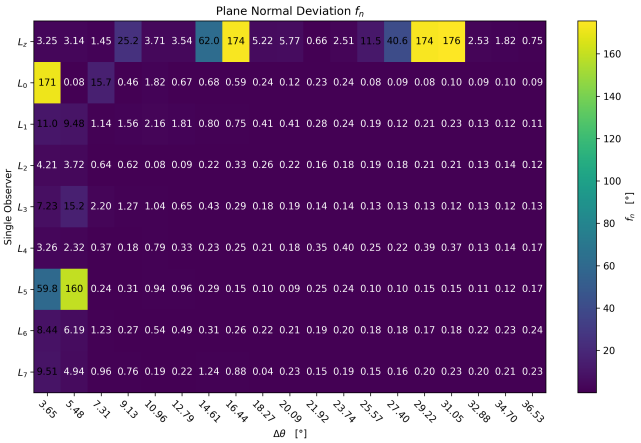


Figure 5: Gaussian method: Plane normal deviation.

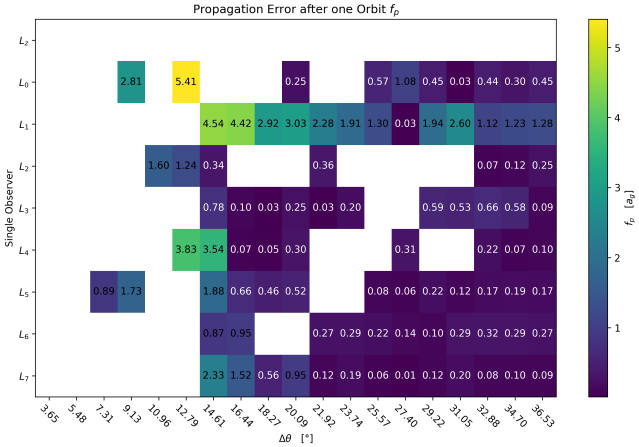


Figure 6: Gaussian method: Propagation error after one orbit.

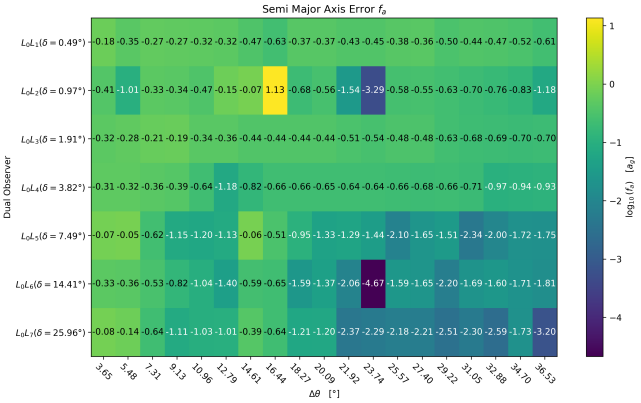


Figure 7: Stereo vision (full parametric model): Stability of semi major axis.

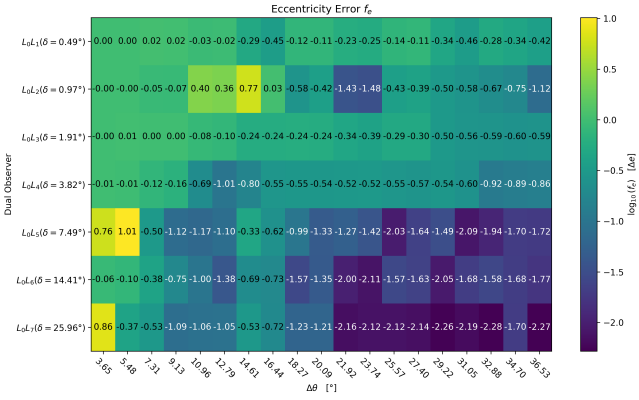


Figure 8: Stereo vision (full parametric model): Stability of eccentricity.



Figure 9: Stereo vision (full parametric model): Plane normal deviation.

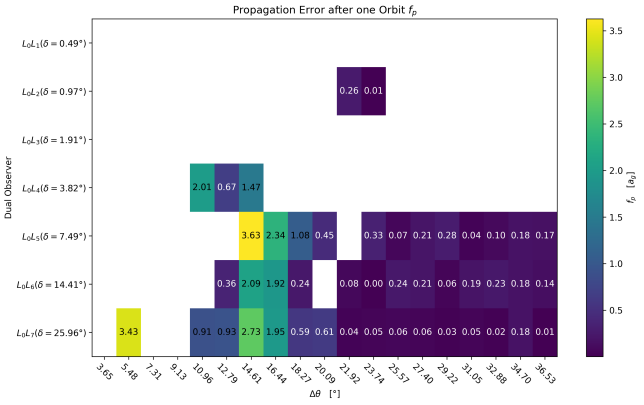


Figure 10: Stereo vision (full parametric model): Propagation error after one orbit.



Figure 11: Simplified Gaussian method: Stability of semi major axis.

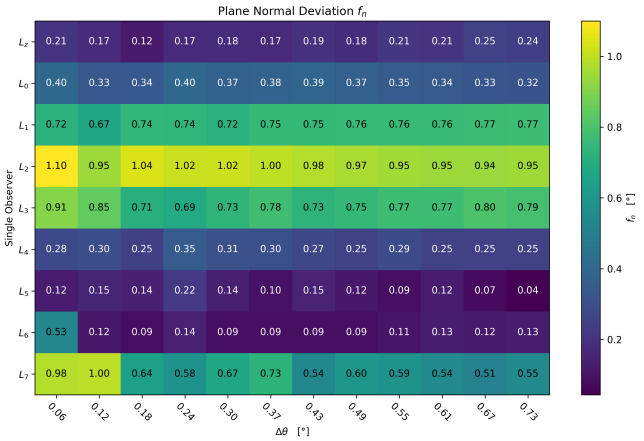


Figure 12: Simplified Gaussian method: Plane normal deviation.

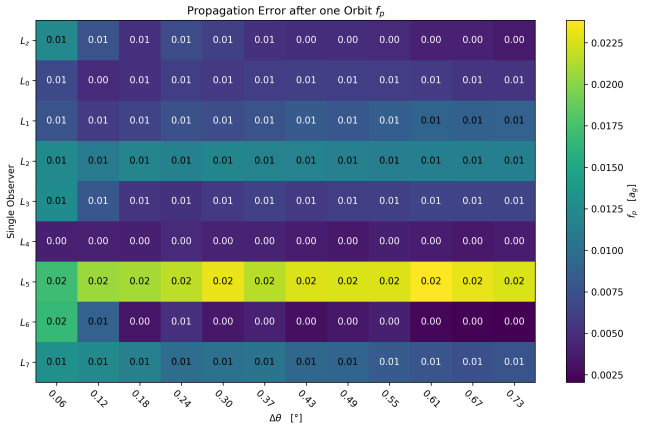


Figure 13: Simplified Gaussian method: Propagation error after one orbit.

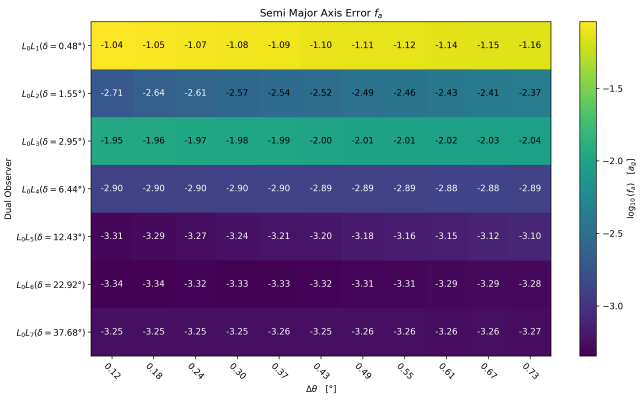


Figure 14: Stereo vision (circular model): Stability of semi major axis.

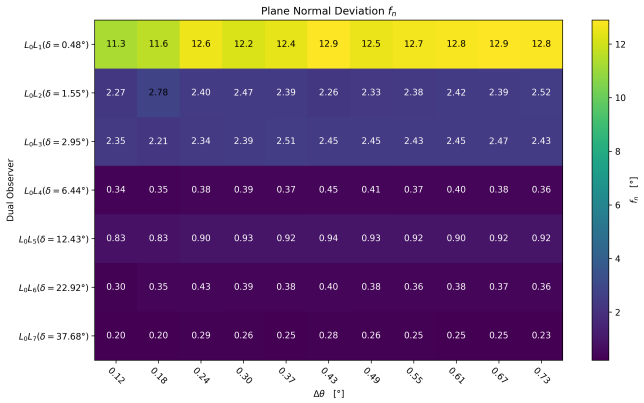


Figure 15: Stereo vision (circular model): Plane normal deviation.

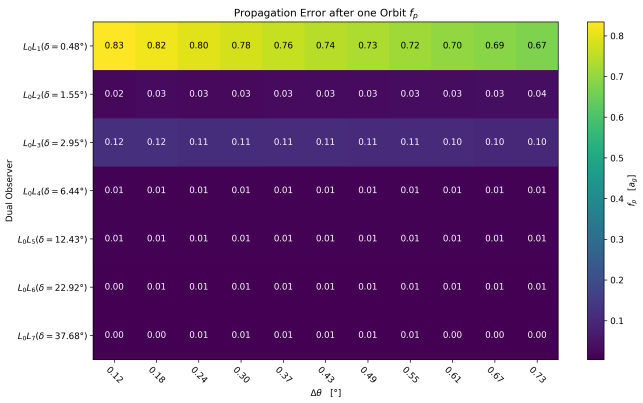


Figure 16: Stereo vision (circular model): Propagation error after one orbit.

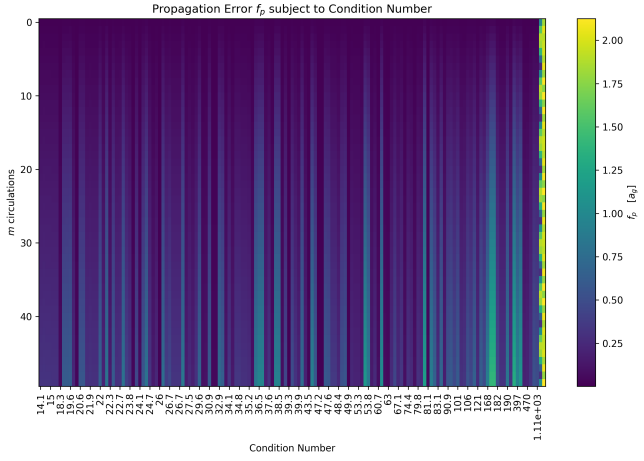


Figure 17: Simplified Gaussian method: Propagation error after m orbits subject to calibration condition number.

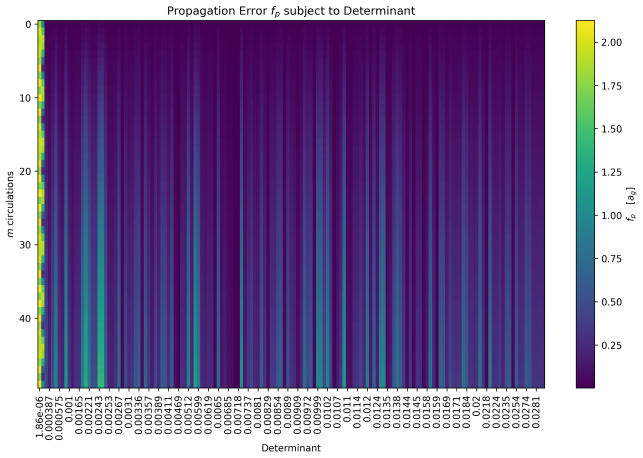


Figure 18: Simplified Gaussian method: Propagation error after m orbits subject to calibration determinant.

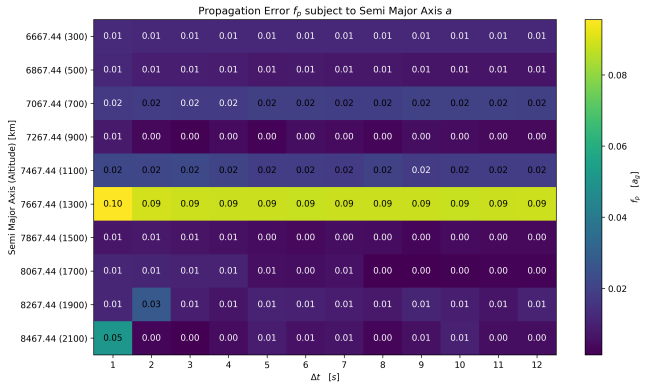


Figure 19: Simplified Gaussian method: Propagation error subject to orbital altitude.

4 Discussion and outlook

The primary goal of this research was to determine how well satellite trajectories are determinable using image-based methods and short observation arcs. Secondary goals consisted of identifying, whether errors in star-based view vector determination have an effect on accuracy and testing if a self developed algorithm for circular orbit determination works for different satellite altitudes.

As for the primary goal, all presented methods were able to predict a satellite's orbit with varying accuracy. The Gaussian and the stereo vision method for full parametric trajectory determination are able to yield reasonably accurate results, for an observation span equating 1/10 of an arc if observers were placed favorably. However, these approaches are unsuitable for high-precision applications as predicted position drift from ground truth steeply, resulting in offsets of tens to hundreds of kilometers after just one circulation.

The simplified Gaussian as well as the stereo vision method for circular trajectory determination are able to yield more accurate results, specifically in terms of parameter accuracy. Especially noteworthy is their required observation span to determine stable parameters which is less than 1/1000 of an arc. The drift between predicted position and ground truth is offset by tens of kilometers after one orbit, so these might be suitable for certain applications.

One aspect that has not been considered in this paper are non-circular ground truth orbits. They may be predictable by the Gaussian and stereo vision method for full parametric trajectory determination, but further research would need to be conducted to determine whether results are of similar accuracy. Models for circular determination would generally be unsuitable for this task, whereby the resulting error between elliptical ground truth and circular prediction may also be of research interest.

As for the secondary objective, we hypothesized that instabilities in star-based view vector determination might yield inaccurate orbital predictions. Results show, that there seems to be no obvious correlation between accuracy and determinant or condition number of the star calibration matrix for general cases. This means that there must be other sources of noise to be investigated. A potential candidate might be the limited camera resolution. In this context, it is also noteworthy to state that *Space Engine* is a proprietary black box. Usually, this simulation is quite a good representation of space, however there might be unknown rendering effects which are not accounted for. A future approach would be to verify, whether predicted accuracies can be replicated in other simulations or even in real life.

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